Assignment 2 Report

Fitts Law- From Chick-Mate_OutputFile.csv

1. Introduction

We used a pinch and a ray cast from the controller to select the spheres. Our spheres were 0.05 for the smaller sphere and 0.15 for a larger sphere. We used distances of 0.5 and 1.0 so the spheres were not too far from the user. We used left, right, Up, and center for directions which can be seen in our experiment. The spheres move around the user's field of view noticeably but not too much as to go out of eye sight. We opted to use 2 repetitions and we asked the user to complete all pinch interactions before moving to the ray.

2. Trial Values

Trial Number	Method	Distance	Size	Direction	Repetition	Error Rate
1	Pinch	0.5	0.05	Center	1	0
2	Pinch	1.0	0.05	Left	1	90
3	Pinch	0.5	0.15	Right	1	0
4	Pinch	1.0	0.15	Up	1	0
5	Pinch	0.5	0.05	Center	2	0
6	Pinch	1.0	0.05	Left	2	0
7	Pinch	0.5	0.15	Right	2	0
8	Pinch	1.0	0.15	Up	2	0
9	Ray	0.5	0.05	Center	1	50
10	Ray	1.0	0.05	Left	1	80
11	Ray	0.5	0.15	Right	1	50

12	Ray	1.0	0.15	Up	1	50
13	Ray	0.5	0.05	Center	2	0
14	Ray	1.0	0.05	Left	2	80
15	Ray	0.5	0.15	Right	2	0
16	Ray	1.0	0.15	Up	2	50

3. Fitt's Law Metrics

We are a group of 3 so we used the Shannon's model provided as well as Welford's Model.

Shannon's Model

$$MT = a + b * log_2(\frac{A}{W} + 1)$$

Welford's Model

$$MT = a + b * log_2(\frac{A}{W} + 0.5)$$

Trial	Op Values	Fitts Law Metric (Shannon's Model	Welford's Model
1	A = 0.5, W = 0.05, MT = 3.263	ID = 3.4594, TP=1.0602	ID = 3.392, TP = 1.039
2	A = 1, W = 0.05, MT=1.198	ID = 4.392, TP=3.666	ID = 4.3576, TP = 3.637
3	A = 0.5, W = 0.15, MT = 0.882	ID = 2.1155, TP = 2.399	ID = 1.9386, TP = 2.198
4	A = 1, W = 0.15, MT = 0.710	ID = 2.9386, TP = 4.1389	ID = 2.8413 TP = 4.002

5	A = 0.5, W = 0.05, MT = 0.960	ID = 3.459, TP = 3.604	ID = 3.392, TP = 3.533
6	A = 1, W = 0.05, MT = 1.239	ID = 4.392, TP = 3.544	ID = 4.3576, TP = 3.517
7	A = 0.5, W = 0.15, MT = 1.016	ID = 2.1155, TP = 2.082	ID = 1.9386, TP = 1.7528
8	A = 1, W = 0.05, MT =1.267	ID = 2.9386, TP = 2.3193	ID = 4.3576, TP = 3.439
9	A = 0.5, W = 0.05, MT = 2.980	ID = 3.459, TP =1.1609	ID = 3.392, TP = 1.128
10	A = 1, W = 0.05, MT = 0.665	ID = 4.392, TP = 6.604	ID = 4.3576, TP = 6.5528
11	A = 0.5, W = 0.15, MT = 0.612	ID = 2.1155, TP = 3.4567	ID = 1.9386, TP = 3.1676
12	A = 1, W = 0.15, MT = 0.640	ID = 2.936, TP = 4.591	ID = 2.8413, TP = 4.439
13	A = 0.5, W = 0.05, MT = 0.919	ID = 3.459, TP = 3.764	ID = 3.392, TP = 3.691
14	A = 1, W = 0.05, MT = 0.766	ID = 4.392, TP = 5.733	ID = 4.3576, TP = 5.689
15	A = 0.5, W = 0.15, MT = 0.751	ID = 2.1155, TP = 2.817	ID = 1.9386, TP = 2.581
16	A = 1, W = 0.15, MT = 0.766	ID = 2.938, TP = 3.8363	ID = 2.8413, TP =3.709

Shannon's Model Averages:

Trial Pair	Average Throughput	Average ID	Average MT
1 and 5	2.3316	3.4594	2.1115

2 and 6	3.605	4.392	1.107
3 and 7	2.2405	2.1155	0.994
4 and 8	1.2287	2.9386	0.9885
9 and 13	2.4626	3.4594	1.9495
10 and 14	6.1685	4.392	0.7155
11 and 15	3.1368	2.1155	0.6815
12 and 16	4.2139	2.9386	0.703

Method	a (Intercept)	b (Slope)	Model
Ray	0.835	0.144	MT = 0.835 + 0.144 * ID
Pinch	0.624	0.120	MT = 0.624 + 0.120 * ID

Welford's Model Averages:

Trial Pair	Average Throughput	Average ID	Average Movement
1 and 5	2.286	3.392	2.1115
2 and 6	3.577	4.3576	1.107
3 and 7	1.9754	1.9386	0.994
4 and 8	3.7205	2.8413	0.9885
9 and 13	2.4095	3.392	1.9495

10 and 14	6.1209	4.3576	0.7155
11 and 15	2.8743	1.9386	0.6815
12 and 16	4.074	2.8413	0.703

Linear Regression

Method	a (Intercept)	b (Slope)	Model
Ray	0.863	0.140	MT = 0.835 + 0.140 * ID
Pinch	0.640	0.119	MT = 0.640 + 0.119 * ID

4. Takeaways / Results

Based on the Fitts' Law models, we observed that the pinch interaction performed better overall compared to the ray interaction. From the linear regression results using Shannon's model, the pinch method had both a lower intercept (a = 0.624) and a lower slope (b = 0.120) than the ray method (a = 0.835, b = 0.144). This indicates that pinch interaction not only had a faster baseline movement time, but also scaled better as the index of difficulty increased. In contrast, the ray interaction required more time to initiate and was more affected by increases in distance and decreases in target size.

When comparing Shannon's model and Welford's model, both produced similar trends in terms of movement time and throughput. However, Shannon's model consistently produced slightly higher ID values due to the +1 term in the formula, which better accounts for smaller amplitudes and is more widely used in human-computer interaction studies. Despite these differences, both models support the same conclusion that pinch interaction is more efficient than ray-based interaction in this experiment.

In terms of performance, the application worked well overall, but the ray interaction showed higher error rates in several trials, especially at larger distances and smaller target sizes. This suggests that aiming with a ray is less precise compared to direct hand interaction. To improve the application, we could increase the target size slightly or add visual feedback such as a cursor or highlight to make ray selection more accurate.